Jinghao Yan

PhD Student · University of Science and Technology of China
Embodied Intelligence | Dexterous Manipulation | Multimodal Perception Fusion
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Education

University of Science and Technology of China Sep 2026 – Present
PhD Student in Electronic Information, Department of Automation

Tongji University Sep 2022 – Jun 2026
Bachelor's Degree in Computer Science and Technology
GPA: 4.97/5.00 | Rank: 1/118 (top 0.85%)

Research Experience

Shanghai AI Lab Oct 2025 – Present
Research Intern

MegaParts: Scaling Part-Aware 3D Object Generation to 300 Parts via Token-Efficient Autoregressive Modeling

SIGGRAPH Asia 2026 · Accepted

Research: Scales part-aware 3D generation to 300 parts using adaptive-length discrete geometry tokens and a unified autoregressive sequence for object bounds, part bounds and geometry. Compact representations reduce long-context costs while preserving fine structures and part-level control.

Contribution: Ran baseline experiments, compiled results and analyzed comparisons to support quality and effectiveness evaluation; gained a solid understanding of the overall framework, part representations and experimental design.

Code-as-Room: Generating 3D Rooms from Top-Down View Images via Agentic Code Synthesis

Preprint

Research: Generates 3D rooms from top-down images by parsing scene elements and spatial relations, then synthesizing Blender code for geometry, materials and lighting. A staged execution framework and cross-stage memory address context forgetting, unstable execution and agent loops.

Contribution: Integrated a 3D generation model into the agent system to create objects difficult to construct directly through code, complementing procedural modeling and broadening the range of objects the system can generate.

From Parts to Sim: Bottom-Up Autoregressive Generation of Sim-Ready 3D Assets

Under review

Research: Generates simulation-ready assets through bottom-up autoregressive modeling of fine-grained components, rigid links, joints and physical properties. Assets support USD/URDF export and provide simulated interaction trajectories for robot policies that transfer to real objects.

Contribution: Ran and analyzed baseline experiments; built the material pipeline using part UV atlases, object-level material generation and per-part remapping to maintain consistent appearance while preserving articulation.

Honors and Awards

National Scholarship (Top 3%)	2022–23
National Scholarship (Top 3%)	2023–24
National Scholarship (Top 3%)	2024–25
Shanghai Outstanding Graduate (Top 2%)	2026

Technical Skills

Programming & frameworks: Python, PyTorch

Deep learning: Model deployment and experimental evaluation

3D tools: Experience with Blender and 3D asset processing